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Design of parallel feed-forward compensator and its application to electromechanical system with friction load

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Abstract: This contribution deals with design and application of parallel feedforward compensator (PFC) for an electromechanical system with friction load. For a prototypic electromechanical system with a nonlinear friction curve and a classical cascade drive control it is shown that self-excited vibrations may occur in closed-loop operations. This phenomenon is connected with the unstable zero dynamics. In order to solve this problem the application of parallel compensation as an extension for conventional control scheme is proposed.

Keywords: Unstable zero dynamics; parallel feed-forward compensator; self excited vibrations; stabilization.

1. INTRODUCTION

Design of a feedback control system for a plant which possesses zeros with positive real parts is always a challenging task, as these systems are known to be unstable under high controller gains. In addition, it is well known that the location of system zeros is invariant with respect to any feedback. On the other hand, using small controller gains results in low performance of closed loop system transients. A classical solution involves design of a complex control law. It increases control system complexity significantly making a practical implementation more expensive and time-consuming.

In this contribution a different, alternative approach allowing the application of standard, simple feedback control will be presented. The closed loop stability can be preserved applying an additional PFC, which results in an augmented system with redefined output. As a first step the parallel interconnection of the plant G and compensator G_{pc} allows to assign the system zeros in the left half plane. In a second step, a standard feedback controller without gain limitations can be applied to stabilize the plant.

There are a lot of contributions being dedicated to the application of parallel compensation: Smith (1958) for delay compensation; Bar-Kana (1987), Kaufman et al. (1998) for rendering a system almost strictly positive real (ASPR) thereafter it can be easily stabilized by output feedback; Deng et al. (1999) for robust stabilization of uncertain systems; Gessing (2004) PFC design for stable plants with unstable zeros and others (Kim et al. (2016), Mizumoto and Tanaka (2010), Isidori and Marconi (2008)).

Although the design approach for stable systems has been studied in Gessing (2004) and applied in Gavini et al. (2011), systematic PFC design for unstable plants with unstable zero dynamics is still not well investigated. A first approach has been presented in Palis and Kienle (2014). In this contribution, a simple design method for a vast class of unstable systems with unstable zero dynamics and its application to a practical problem is proposed.

A typical example for closed-loop instabilities caused by unstable zero dynamics are electromechanical systems, where the system operates at a region of negative slope of its nonlinear friction-torque curve (Farhang and Lim (2006), Rahmani et al. (2015)). In this case self-excited oscillations may occur (Klepikov (2014), Fidlin (2006), Hess (1958)). Most commonly these oscillations decrease the operational performance and can lead to mechanical breakdowns. Their occurrence has been reported in different areas, e.g. rail transport, metal-working machinery, bridge cranes, rolling mills, industrial robots, drilling systems etc. (Klepikov (2014), Wang et al. (2013), Kiseleva et al. (2015)).

This contribution is structured as follows: in section 2 the general PFC design procedure is presented. In section 3 the main problems applying standard control approaches in electromechanical systems are discussed and the mathematical model of a two degree-of-freedom (DoF) electromechanical system with a nonlinear friction curve is presented. In section 4 the design of an appropriate parallel compensator for the presented electromechanical system is described.

2. PFC DESIGN PROCEDURE

Consider the following single-input single-output (SISO) linear time-invariant (LTI) plant, which may contain left-half plane (LHP) as well as right-half plane (RHP) poles and zeros:

$$G(s) = \frac{N^{+}(s)N^{-}(s)}{D^{+}(s)D^{-}(s)}, \qquad (1)$$

where $N^{+}(s)$ and $D^{+}(s)$ are polynomial fractions with LHP roots, $N^{-}(s)$ and $D^{-}(s)$ fractions with RHP roots. In order to influence the stability behaviour of zero dynamics an appropriate PFC $G_{pc}(s)$ as depicted in Fig. 1

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has to be designed.

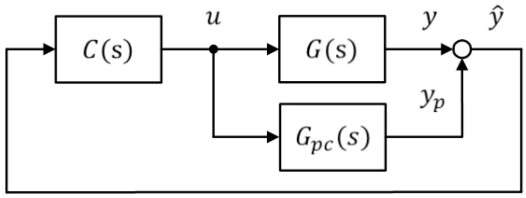

Fig. 1. Overall control

It is assumed that the PFC consists of the fractions containing all plant poles and zeros with negative real parts, i.e. $N^+(s)$ and $D^+(s)$, plus two additional fractions $N_c(s)$ and $D_c(s)$ that can be assigned:

$$G_{pc}(s) = \frac{N^+(s)N_c(s)}{D^+(s)D_c(s)}. \tag{2}$$

The parallel interconnection of the plant and the PFC results in

$$G_s(s) = \frac{N^+(s)\left[N^-(s)D_c(s) + N_c(s)D^-(s)\right]}{D^+(s)D^-(s)D_c(s)}. \tag{3}$$

Here, the stability conditions of zero dynamics for the augmented system $G_s(s)$ can be derived if and only if the numerator fraction $N^-(s)D_c(s)+N_c(s)D^-(s)$ has no RHP roots. This fraction can be reformulated as:

$$1 + \frac{D_c(s)N^-(s)}{N_c(s)D^-(s)} = 0. \tag{4}$$

Thus, the design of a parallel compensator which stabilizes the unstable zero dynamics is equivalent to designing a controller

$$G_c(s) = \frac{D_c(s)}{N_c(s)} \tag{5}$$

which stabilizes the virtual plant consisting of the system RHP poles and zeros

$$G^-(s) = \frac{N^-(s)}{D^-(s)}. \tag{6}$$

Thereafter, this controller $G_c(s)$ will be inversely included in the PFC according to eq. 2.
Assuming that $D^-(s)$ and $N^-(s)$ are coprime polynomials the closed loop system will be internally stable and all system sensitivity functions will contain only desirable poles if and only if G_c is parametrized as follows

$$G_c(s) = \frac{\dfrac{D_c(s)}{L(s)} + Q(s)\dfrac{D^-(s)}{L(s)}}{\dfrac{N_c(s)}{L(s)} - Q(s)\dfrac{N^-(s)}{L(s)}}, \tag{7}$$

where $Q(s)$ is a proper stable transfer function that contains only desirable poles, $N_c(s)$ and $D_c(s)$ are polynomials that satisfy the following pole placement equation

$$L(s) = N^-(s)D_c(s) + D^-(s)N_c(s), \tag{8}$$

where $L(s)$ is some arbitrary chosen polynomial of suitable degree, which contains zeros in desirable complex plane region (Goodwin et al. (2001), Doyle et al. (1996)). To solve the linear Diophantine eq. 8 an approach based on the Sylvester's matrix can be applied.
Let assume that $D^-(s)$ and $N^-(s)$ are plant polynomials of degree n, $N_c(s)$ and $D_c(s)$ are controller polynomials of degree $m = n-1$, $L(s)$ is some polynomial with desired zeros of degree $k = n+m$, i.e.

$$N^-(s) = a_n s^n + a_{n-1}s^{n-1} + \cdots + a_0, \tag{9}$$
$$D^-(s) = b_n s^n + b_{n-1}s^{n-1} + \cdots + b_0. \tag{10}$$
$$N_c(s) = c_m s^m + c_{m-1}s^{m-1} + \cdots + c_0, \tag{11}$$
$$D_c(s) = d_m s^m + d_{m-1}s^{m-1} + \cdots + d_0, \tag{12}$$
$$L(s) = s^k + l_{k-1}s^{k-1} + \cdots + l_0. \tag{13}$$

The pole placement problem is leading to the solution of the corresponding matrix equation $Mv = x$, where

$$M = \begin{bmatrix} a_n & 0 & \cdots & 0 & b_n & 0 & \cdots & 0 \\ a_{n-1} & a_n & \cdots & 0 & b_{n-1} & b_n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_0 & a_1 & \cdots & a_n & b_0 & b_1 & \cdots & b_n \\ 0 & a_0 & \cdots & a_{n-1} & 0 & b_0 & \cdots & b_{n-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_0 & 0 & 0 & \cdots & b_0 \end{bmatrix},$$

$$v = \begin{bmatrix} d_m \\ \vdots \\ d_0 \\ c_m \\ \vdots \\ c_0 \end{bmatrix}, x = \begin{bmatrix} 1 \\ l_{k-1} \\ \vdots \\ l_0 \end{bmatrix}.$$

According to the Sylvester's theorem matrix M is non-singular if and only if polynomials $N^-(s)$ and $D^-(s)$ are coprime.

3. ELECTROMECHANICAL SYSTEM WITH FRICTION LOADS

3.1 Problem statement

Consider a prototypic two DoF electromechanical system with elastic coupling and frictional load as depicted in Fig. 2. Here, the first mass is actuated by the DC motor with separately excited field winding, whereas friction loads are acting on the second mass, only.

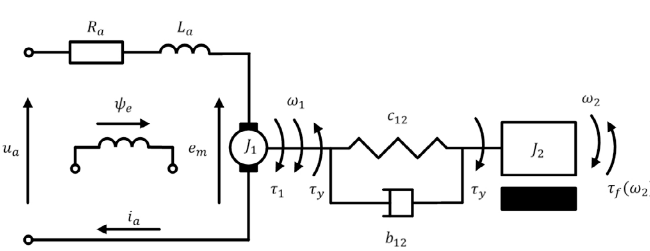

Fig. 2. Electromechanical system with friction load

Nowadays, controlled electric drives have become a quasi-standard for the actuation electromechanical systems. In order to govern the electric energy flow, these drive systems are equipped with power converters based on different semiconductor schemes. In most cases manufacturer offer a complex product augmenting its converter by a fully-parametrizable classical control system and a manufacture-specific software solution. Although, such a design allows fast engineering solutions in many application cases, implementation of non-standard control laws by the user becomes a difficult task.

Velocity control of the electric drive is one of the classical tasks for many machines and mechanisms. For this purpose the cascade control with PI controllers finds a vast practical application due to its high robustness and simplicity in tuning (Fig. 3). It consists of an inner current (or sometimes torque) control loop $G_i(s)$ and an outer velocity control loop $G_\omega(s)$.

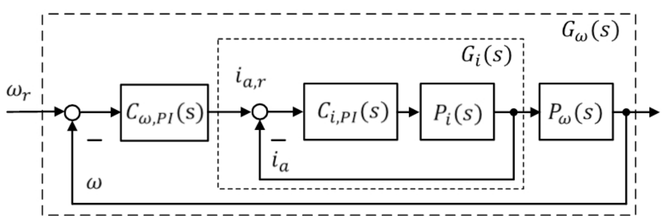

Fig. 3. Standard cascade control

This conventional control scheme is widely used by engineering companies. The velocity measurements in electromechanical systems are often provided with incremental or optical sensors on the motor shaft, as direct velocity measurements on the working element of the mechanisms increase costs and are often complicated or in some cases even impossible realize.

From a theoretical point of view, this means that in most cases only the first mass velocity of electromechanical systems with possibly more than one DoF can be measured. Hence, e.g. due to the elastic couplings to working element this system part is not directly measured and associate eigenvalues result in system zeros (Isidori (2013)). Due to the fact that under certain conditions the system may operate in a negative slope region of the friction load curve problems connected to the unstable zero dynamics may occur.

3.2 Modeling

As a simple simulation model a electromechanical system with two DoF and nonlinear friction curve is studied here. The equations of motion for the mechanical subsystem can be represented as follows:

$$J_1 \frac{d\omega_1(t)}{dt} = \tau_1(t) - \tau_y(t), \tag{14}$$

$$\tau_y(t) = c_{12} \int_0^t (\omega_1(\tau) - \omega_2(\tau))d\tau + b_{12}(\omega_1(t) - \omega_2(t)), \tag{15}$$

$$J_2 \frac{d\omega_2(t)}{dt} = \tau_y(t) - \omega_2(t)\tau_f(\omega_2), \tag{16}$$

where J_1 and J_2 are the inertial masses of the drive system and the working element, $\omega_1(t)$ and $\omega_2(t)$ are the angular velocities of both masses, c_{12} and b_{12} are the torsional stiffness and damping of the linear spring connection between both masses, $\tau_1(t)$ is the motor drive torque, $\tau_y(t)$ is the elastic torque and $\tau_f(\omega_2)$ is the friction load torque function, which depends on the angular velocity of the second mass.

The equation of motion for the electrical subsystem is given by:

$$u_a(t) = e_m(t) + R_a i_a(t) + L_a \frac{di_a(t)}{dt}, \tag{17}$$

where $u_a(t)$ and $i_a(t)$ are the armature voltage and current, R_a and L_a are the armature resistance and inductance and $e_m(t)$ is the electromotive force.

Assuming a fixed magnetic field, the electrical and mechanical system variables can be coupled as:

$$\omega_1(t) = \frac{1}{k_m \psi_e} e_m(t), \tag{18}$$

$$\tau_1(t) = k_m \psi_e i_a(t), \tag{19}$$

where k_m is the DC motor ratio and ψ_e is the constant magnetic field flux.

Modeling of friction loads for a particular system is typically a complicated task, due to limited understanding of the friction phenomenon at hand. It often results in high uncertainties and time varying model quality. However, for our study the presence of a negative slope region in the friction curve is sufficient to result in the occurrence of instabilities and limit cycles (Mihajlovi (2005), Marquez et al. (2015), Puebla and Alvarez-Ramirez (2008)). That is why a simplified humped friction model with the Stribeck effect can be used:

$$\tau_f(\omega_2(t)) = \frac{2a_1 a_2 \omega_2(t)}{\omega_2^2(t) + a_2^2} + a_3\omega_2, \tag{20}$$

where a_1, a_2 and a_3 - are the positive friction curve parameters (Fig. 4).

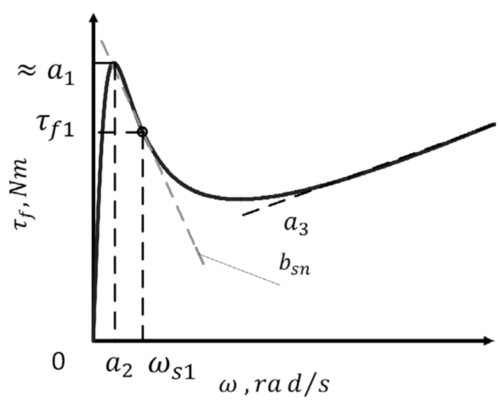

Fig. 4. Friction curve

Applying the Laplace transformation to the system linearised at some negative slope region, the above modeling equations 14, 15, 16 and 17 can be represented as follows

$$\omega_1(s) = \frac{1}{J_1 s}(\tau_1(s) - \tau_y(s)), \tag{21}$$

$$\tau_y(s) = \frac{c_{12}}{s}(\omega_1(s) - \omega_2(s)) + b_{12}(\omega_1(s) - \omega_2(s)), \tag{22}$$

$$\omega_2(s) = \frac{1}{J_2 s}(\tau_y(s) - \omega_2(s)b_{sn}), \tag{23}$$

$$i_a(s) = \frac{1/R_a}{T_a s + 1}(u_a(s) - e_m(s)). \tag{24}$$

where T_a - is the time constant of the electric motor circuit, b_{sn} - is the slope parameter of tangent at the operating

point ω_{s1}.
As has been mentioned, the source of instability in the given model is, from a mathematical point of view, the occurrence of the negative slope b_{sn} for specific regions of the friction velocity curve (Fig. 4). This phenomenon, i.e. the decrease of the friction torque for an increased velocity, results in a further acceleration assuming a constant motor torque. In practical applications negative slope of the friction curve occurs only in a limited region and would therefore prevent the system from speeding up infinitely. Nevertheless, this mechanism of instability may result in limit cycles, i.e. nonlinear oscillations with high amplitudes, which is a frequently observed phenomenon leading to decreased system performance and damages.

3.3 Conventional control system

Design of the classical cascade control system begins with the inner control loop, i.e. the controller $C_{i,PI}(s)$ for the motor current plant $P_i(s)$ is designed first, after that the outer controller $C_{\omega,PI}(s)$ for the whole plant $G_i(s)$ and $P_\omega(s)$ is designed (Fig. 3).
The electrical subsystem $P_i(s)$ can be represented by two first order systems, the rectifier associated with the time constant T_μ and the electrical part of the motor with the time constant T_a (eq. 24), where $T_\mu \ll T_a$. For the current controller a standard PI controller as depicted in Fig. 5 is chosen.

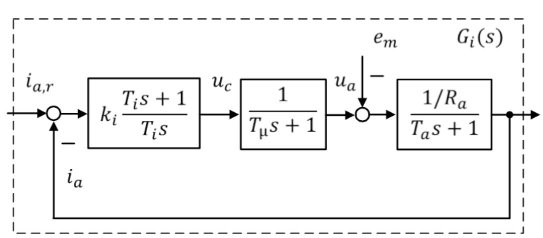

Fig. 5. Current control loop

Here, $u_c(s)$ is the controller voltage, k_i and T_i are the PI-controller parameters.
Applying the magnitude (modulus) optimum tuning procedure as a standard controller design procedure for electric drive systems PI-controller parameters are given by $k_i = T_a R_a / 2T_\mu$ and $T_i = T_a$ (Bajracharya et al. (2008)). Here, the main idea is to compensate the largest time constant T_a in order to achieve fast reference tracking and to provide a good disturbance rejection. Thus, we assume that the electromotive force e_m is compensated by the inner PI controller. Using this tuning the current closed loop system can be approximated as

$$G_i(s) = \frac{i_a(s)}{i_{a,r}(s)} = \frac{1}{2T_\mu s + 1}. \tag{25}$$

In a second step, the outer controller is designed for the closed inner loop $G_i(s)$ and the mechanical subsystem, which is described by the inertial mass of the overall mechanism $J_s = J_1 + J_2$ (Fig. 6). In order to achieve both good reference tracking and disturbance rejection the parameters of the velocity PI-controller and the reference prefilter can be adjusted according to the symmetrical optimum, i.e. $k_\omega = J_s / 4T_\mu k_m \psi_e$, $T_\omega = 8T_\mu$ and $T_v = 8T_\mu$.

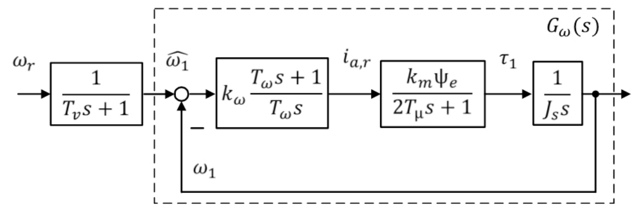

Fig. 6. Velocity control loop

However, as has been mentioned above, velocity measurements can not be performed on the working element of the system. Hence, some system part is not directly measured (Fig. 7). Assuming the damping of the spring connection is very low, i.e. $b_{12} \approx 0$, the transfer function of mechanical subsystem considering the motor torque $\tau_1(s)$ as input and the motor velocity $\omega_1(s)$ as output of interest can be expressed as follows.

$$P_\omega(s) = \frac{J_2 s^2 + b_{sn} s + c_{12}}{J_1 J_2 s^3 + J_1 b_{sn} s^2 + (J_1 + J_2) c_{12} s + b_{sn} c_{12}} \tag{26}$$

Investigating the numerator fraction of the eq. 26, the problem of unstable zero dynamics arises for slope coefficients $b_{sn} < 0$.
In order to analyse the stability properties of linearised system with conventional velocity control (Fig. 7), the root locus depicted in Fig. 8 has been derived considering varying gain k_ω and fixed integrator time constant T_ω tuned to symmetrical optimum. The associate parameter values are given in Tab. 1. As can be observed in Fig. 8, the PI controller can stabilize the system dynamics only for a small range of the controller gain $0.095 < k_\omega < 0.38$ (black line) due to the presence of RHP zeros.
It should be emphasized that this instability is a local effect for the described linear system resulting globally in nonlinear oscillations. Taking into account the nonlinear friction curve (eq. 20) and the aforementioned tuning of the PI controller the step response of the nonlinear system is represented in Fig. 9. As can be seen, the aforementioned unstable zero dynamics lead to system instability and the occurrence of nonlinear oscillations on the second mass whereas the first mass is stabilized by PI controller. It should be mentioned that depending on the mass relations oscillations of the second mass may be imperceptible in real measurements.

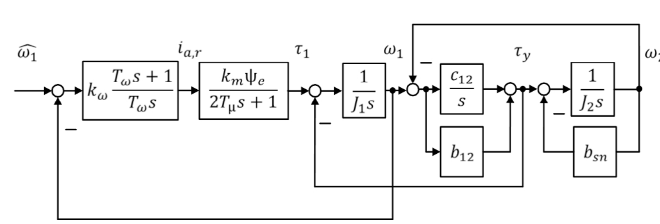

Fig. 7. Velocity control loop with additional dynamics

4. DESIGN OF A PFC

Taking into account the aforementioned nominal parameters the plant transfer function consisting of the current

Table 1. Simulation parameters

T_a	0.1	$[s]$
T_μ	0.005	$[s]$
R_a	1	$[\Omega]$
J_1	0.5	$[kg\ m^2]$
J_2	0.1	$[kg\ m^2]$
c_{12}	1	$[Nm/rad]$
b_{12}	0	$[Nm\ s/rad]$
$k_m\psi_e$	1	$[rad/s]$
a_1	0.085	$[-]$
a_2	2	$[-]$
a_3	0.0015	$[-]$
ω_{s1}	5	$[rad/s]$
b_{sn}	-0.007	$[-]$

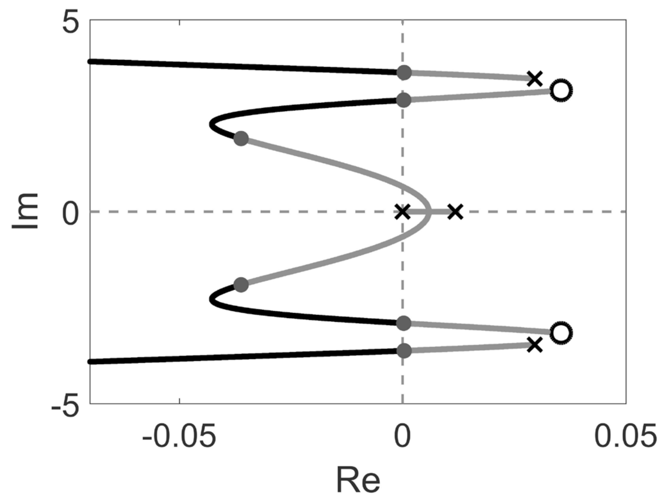

Fig. 8. Root locus for system with unstable zero dynamics

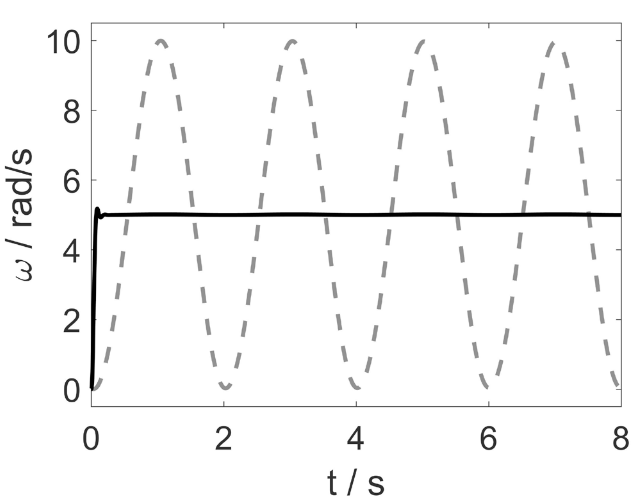

Fig. 9. Angular velocities of the first (solid black) and second (dotted grey) masses

closed loop system $G_i(s)$ and mechanical subsystem $P_\omega(s)$ can be calculated as:

$$G(s) = \frac{200(s^2 - 0.07061s + 10)}{(s+100)(s-0.01177)(s^2-0.05884s+12)}. \quad (27)$$

Here, the system can be separated into the following stable and unstable fractions:

$$N^-(s) = (s^2 - 0.07061s + 10), \quad (28)$$

$$D^-(s) = (s - 0.01177)(s^2 - 0.05884s + 12), \quad (29)$$

$$N^+(s) = 200, \quad (30)$$

$$D^+(s) = (s + 100). \quad (31)$$

In order to achieve stable zero dynamics the following polynomial should have no RHP zeros:

$$1 + G_c(s)G^-(s) = 0. \quad (32)$$

To stabilize this system the pole placement approach utilizing polynomial methods has been applied. The characteristic polynomial $L(s)$ eq. 13 can be represented as follows:

$$L(s) = s^k + \lambda_{k-1}\omega_0 s^{k-1} + \lambda_{k-2}\omega_0^2 s^{k-2} + \cdots + \omega_0^k, \quad (33)$$

where $\lambda_i = l_i/\omega^{n-i}$ are normalized coefficients, i.e. $\omega_0 = \sqrt[k]{l_0}$.

The desired poles location has been chosen according to the reverse Bessel polynomial of degree $k = 5$ (Grosswald (1978)) for $\omega_0 = 2$.

$$L(s) = s^5 + 7.62s^4 + 27.11s^3 + 55.09s^2 + 62.98s + 32 \quad (34)$$

Solving the matrix equation $Mv = x$ we receive polynomial coefficients resulting in stabilizing controller for the virtual plant.

$$G_c(s) = \frac{19.249(s+3.239)(s+0.04057)}{(s-14.77)(s+3.214)} \quad (35)$$

Considering equation 2 the parallel compensator can thus be represented as follows

$$G_{pc}(s) = \frac{10.39(s-14.77)(s+3.214)}{(s+100)(s+3.239)(s+0.04057)}. \quad (36)$$

Applying the designed compensator $G_{pc}(s)$ in parallel connection with the transfer function $G(s)$ yields an augmented system $G_s(s)$ with locally asymptotically stable zero dynamics. To analyse the stability properties of this augmented system with the aforementioned velocity feedback control, the root locus for the varying gain k_ω (Fig. 10) has been calculated.

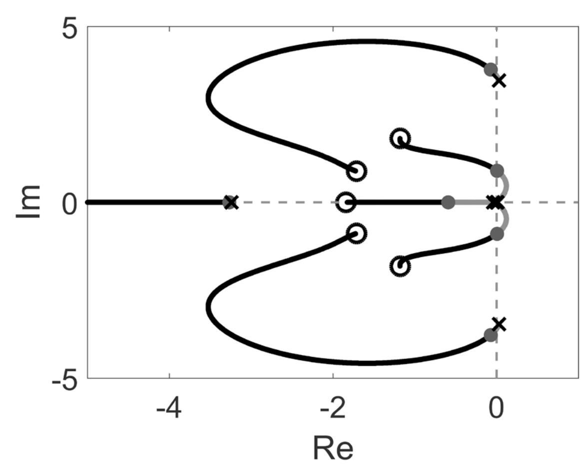

Fig. 10. Root locus for augmented system

In this specific example no additional controller retuning is needed and hence the above presented tuning of PI-controller can be used. However, to increase the overall system performance $G_{cl} = \omega_1(s)/\hat{\omega_1}(s)$ the system zero at $s = -0.04512$ can be compensated by re-tuning the prefilter time constant $T_v = 24.64$. Step responses for both masses are depicted in Fig. 11.

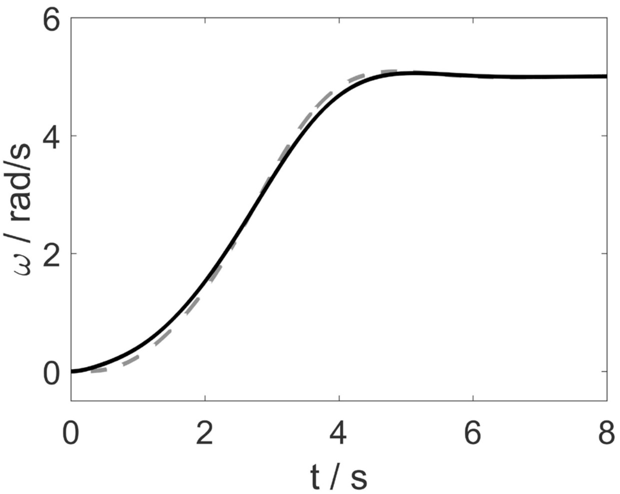

Fig. 11. Angular velocities of the first (solid black) and second (dotted grey) masses

5. CONCLUSION

In this contribution we proposed a PFC design procedure for an unstable plant with unstable zero dynamics. The parallel interconnection of the PFC and the plant results in a new system with asymptotically stable zero dynamics that allows to use standard feedback control systems without gain limitations. It has been shown that this approach can be successfully applied to electromechanical system with friction loads.

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